POWER PULSE

Overhead Line Insulators Important Formulae

Notes by Balaji Sir

1 Voltage distribution in a string

1. $k = \dfrac{C_{shunt}}{C_{self}}$ (pin-to-earth/self capacitance), unit next to the cross-arm $= V_1$

2. $$V_2 = V_1(1 + k), \qquad V_3 = V_1(1 + 3k + k^2)$$

3-disc string

3. $$V_4 = V_1(1 + 6k + 5k^2 + k^3)$$

4-disc string

4. $$V = V_1 + V_2 + V_3 = V_1(3 + 4k + k^2)$$

(3 discs)

5. $$\text{String efficiency} = \frac{V_{string}}{n \times V_{\text{disc nearest the conductor}}} \times 100$$

always < 100%

6. $$n = \frac{V_{ph}}{V_{disc} \times \eta_{string}}$$

number of discs

2 Improving the string efficiency

7. Longer cross-arm ($k \downarrow$), grading ring, capacitance grading

the disc nearest the line has the largest voltage

8. Grading ring: adds capacitance from the metal fittings to the line

3 Insulator ratings and types

9. $$\text{Safety factor} = \frac{\text{puncture voltage}}{\text{flash-over voltage}}$$

puncture strength should exceed flash-over

10. Wet flash-over $\approx 0.8 \times$ dry flash-over

11. Pin: up to 33 kV; suspension: above 33 kV; strain: dead ends; shackle: LV

12. Number of discs (approx.): 11 kV–1, 33 kV–3, 66 kV–5, 132 kV–9, 220 kV–14

typical values